



Let the mind manage the body  
Que l'esprit gère le corps

**SCIENCE**

(Subject code No. P141/1)

Index Number: .....

**MAURITIUS EXAMINATIONS SYNDICATE**  
**Primary School Achievement Certificate Assessment**  
**Grade 5 Modular**  
**October 2019**  
**Time: 1 hour** **Total Marks: 50**

**INSTRUCTIONS TO CANDIDATES**

1. Check that this assessment booklet contains 6 questions printed on 10 pages numbered 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11.
2. Write your Index Number on the assessment booklet in the space provided above.
3. You should not use red, green or black ink in answering questions.
4. Write all your answers clearly in the assessment booklet.
5. Attempt all questions.

| Question  | Marking |     | Revision |     | Control |     |
|-----------|---------|-----|----------|-----|---------|-----|
|           | Marks   | Sig | Marks    | Sig | Marks   | Sig |
| 1         |         |     |          |     |         |     |
| 2         |         |     |          |     |         |     |
| 3         |         |     |          |     |         |     |
| 4         |         |     |          |     |         |     |
| 5         |         |     |          |     |         |     |
| 6         |         |     |          |     |         |     |
| Total     |         |     |          |     |         |     |
| Sig (HoG) |         |     |          |     |         |     |

**QUESTION 1 (5 marks)**

Circle the letter which shows the correct answer.

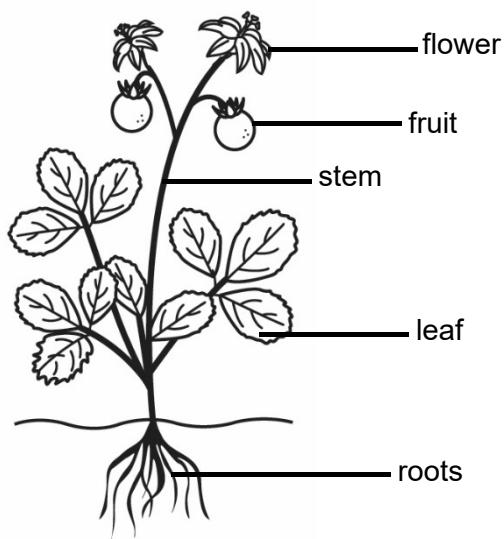
1. Which one of the following is in the **solid** state?

- A** Air
- B** Ice
- C** Steam
- D** Water

2. **Diagram 1** shows a plant.

Which part of the plant carries water from the roots to its other parts?

- A** Flower
- B** Fruit
- C** Leaf
- D** Stem



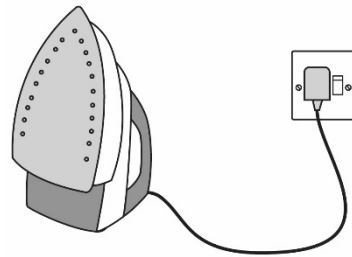
**Diagram 1:** A plant

3. Which of the following helps to **prevent** soil erosion?

- |                                      |                                 |
|--------------------------------------|---------------------------------|
| <b>A</b> Burning of forests          | <b>B</b> Cutting down of trees  |
| <b>C</b> Growing of dense vegetation | <b>D</b> Overgrazing by animals |

4. Which **main** energy transformation takes place in an iron when it is in use?

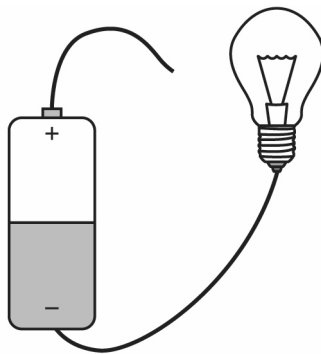
- A** Electrical energy to heat energy
- B** Electrical energy to sound energy
- C** Light energy to electrical energy
- D** Light energy to heat energy



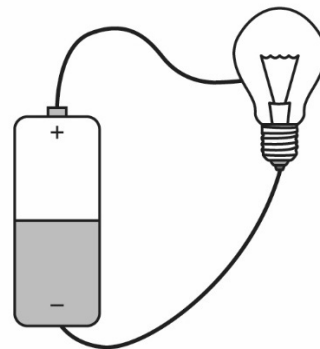
**Diagram 2:** An iron

5. In which one of the circuits below does the bulb light up?

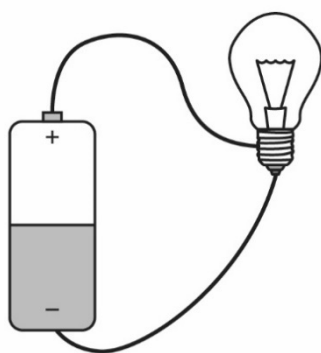
**A**



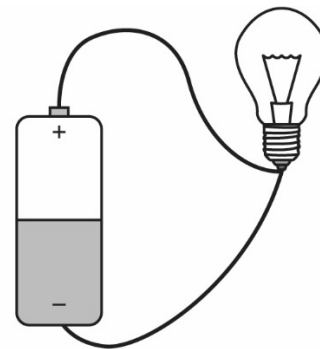
**B**



**C**



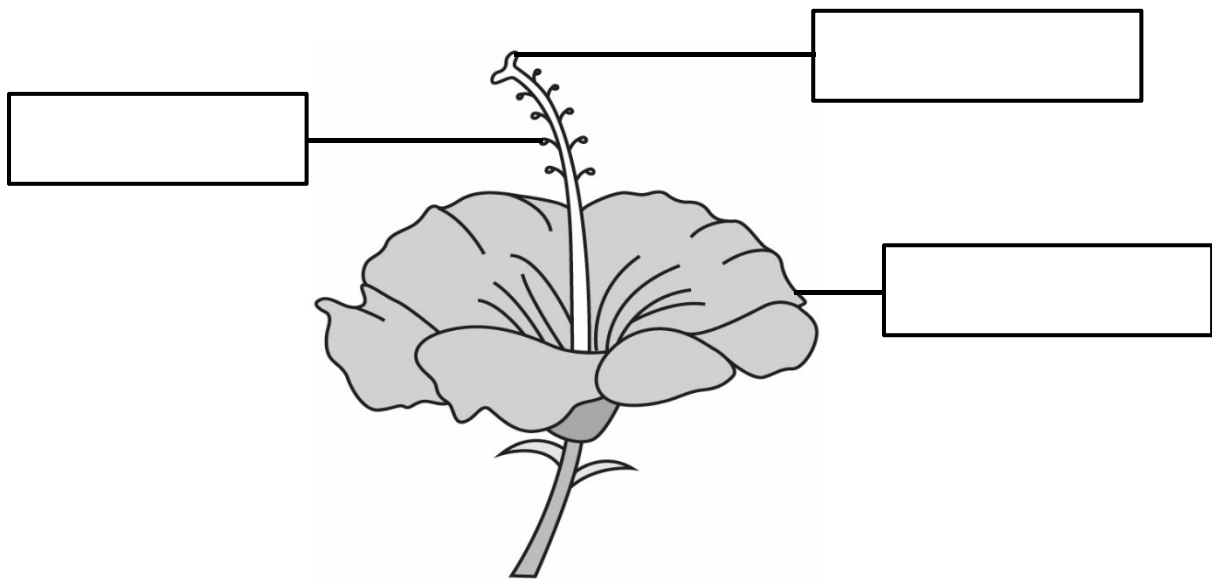
**D**



## QUESTION 2 (10 Marks)

Marks

(a) **Diagram 3** shows a hibiscus flower.



**Diagram 3:** A hibiscus flower

(i) Label the parts of the flower in the boxes in **Diagram 3**. [3]

(ii) Why are flowers usually **colourful**?

\_\_\_\_\_ [1]

(iii) Describe what happens during **pollination**.

\_\_\_\_\_  
\_\_\_\_\_ [2]

(b) **Fill in the blanks** with the correct words from the list below.

|                       |               |               |               |                 |              |
|-----------------------|---------------|---------------|---------------|-----------------|--------------|
| <b>carbon dioxide</b> | <b>leaves</b> | <b>oxygen</b> | <b>petals</b> | <b>sunlight</b> | <b>water</b> |
|-----------------------|---------------|---------------|---------------|-----------------|--------------|

Plants make their food in the presence of \_\_\_\_\_ and a green-coloured substance found in \_\_\_\_\_. They also need \_\_\_\_\_ which is taken up from the soil, and \_\_\_\_\_ which is present in the air. [4]

**QUESTION 3 (8 Marks)**

**Marks**

- (a) **Diagram 4** shows four animals which live in different habitats.



**Fish**



**Earthworm**



**Bird**



**Frog**

**Diagram 4:** Animals living in different habitats

- (i) Choose an animal from **Diagram 4** that lives

1. in the soil. \_\_\_\_\_ [1]

2. **both** in water and on land. \_\_\_\_\_ [1]

- (ii) Some animals live on land and lay their eggs in water.

How do you call this type of animals?

\_\_\_\_\_ [1]

- (iii) A **habitat** is a place where animals live.

Give **two** reasons why a habitat is important for animals.

1. \_\_\_\_\_

2. \_\_\_\_\_

[2]

(b) **Diagram 5** below shows a Kestrel which is endemic to Mauritius.

**Marks**



**Diagram 5:** A Kestrel

(i) What is an **endemic** bird?

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[1]

(ii) Name a bird which is endemic to **Rodrigues**.

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[1]

(iii) It is important to protect **endangered** birds. Give **one** reason why.

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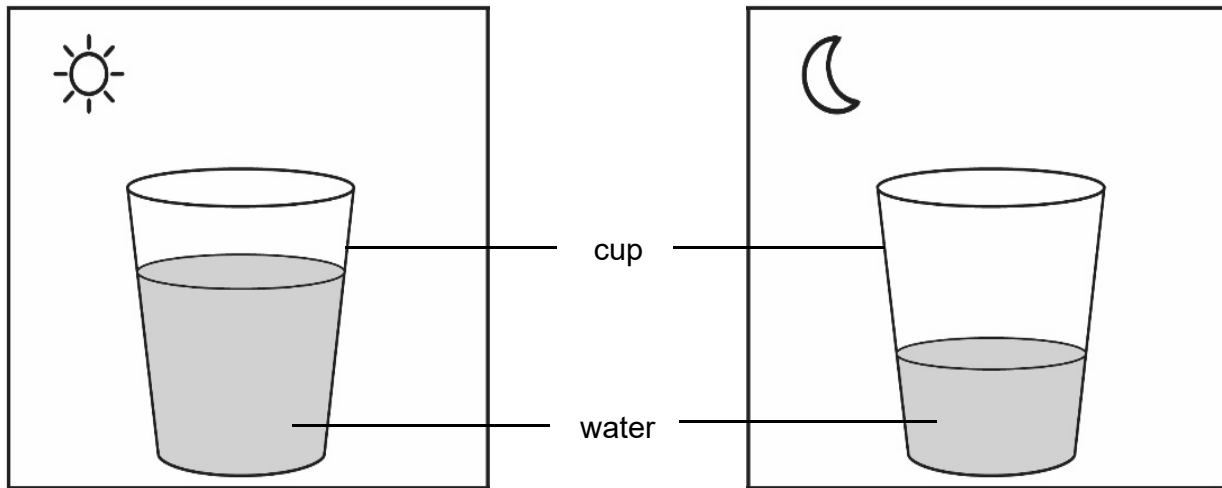
[1]

**QUESTION 4 (7 Marks)**

**Marks**

- (a) Sanjay places a cup of water in the sun in the morning as shown in **Diagram 6A**.

In the evening, he sees that the level of water has decreased as shown in **Diagram 6B**.



**Diagram 6A:** In the morning

**Diagram 6B:** In the evening

- (i) Name the **process** which has caused the level of water to decrease.

\_\_\_\_\_

[1]

- (ii) What could Sanjay have done to prevent the process in **part (i)** from happening?

\_\_\_\_\_

[1]

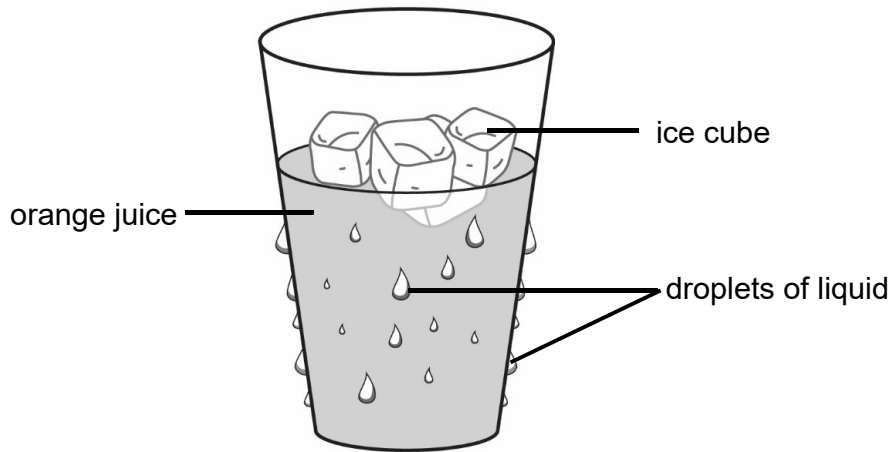
- (b) State two **forms** of energy obtained from the sun.

1. \_\_\_\_\_

2. \_\_\_\_\_

[2]

- (c) Ice cubes are added to a glass containing orange juice.  
After some time, droplets of a liquid are formed on the **outer surface** of the glass.



- (i) Name the **process** by which these droplets are formed.

\_\_\_\_\_ [1]

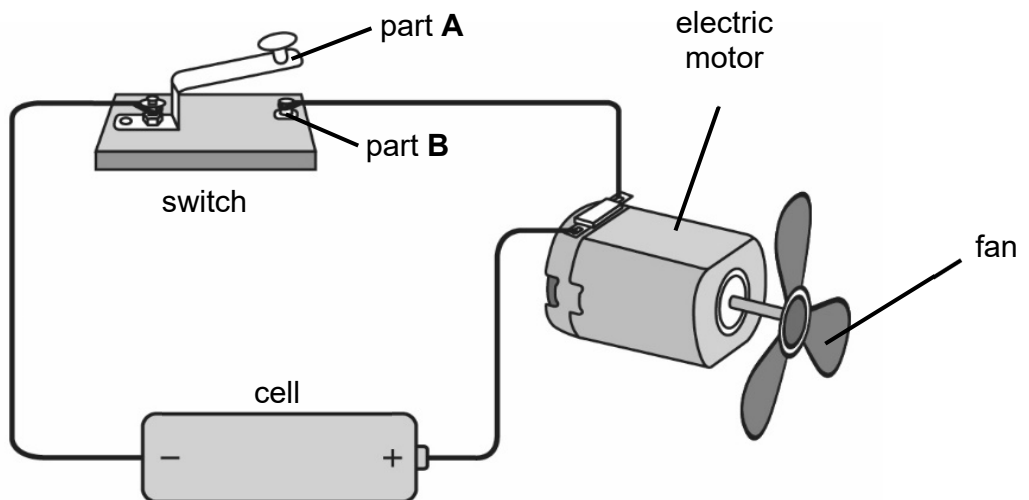
- (ii) Explain how these droplets are formed.

\_\_\_\_\_  
\_\_\_\_\_ [2]

### QUESTION 5 (10 Marks)

**Diagram 7** shows an electric motor connected in a circuit.

In the switch, there is a gap between part **A** and part **B**.



**Diagram 7:** A circuit



(a) (i) Is the circuit in **Diagram 7** closed or open? Explain your answer.

Marks

\_\_\_\_\_

\_\_\_\_\_

[2]

(ii) What is the **source** of energy in the circuit?

\_\_\_\_\_

[1]

(b) (i) Suggest an **object** that can be used to fill the gap between parts **A** and **B** to make the fan turn.

\_\_\_\_\_

[1]

(ii) What **main** energy transformations take place when the fan turns?

\_\_\_\_\_ → **electrical energy** → \_\_\_\_\_

[2]

(c) (i) Give **two** ways how you can save energy.

1. \_\_\_\_\_

2. \_\_\_\_\_

[2]

(ii) Give **two** precautions you need to take when using electricity at home.

1. \_\_\_\_\_

2. \_\_\_\_\_

[2]

### QUESTION 6 (10 Marks)

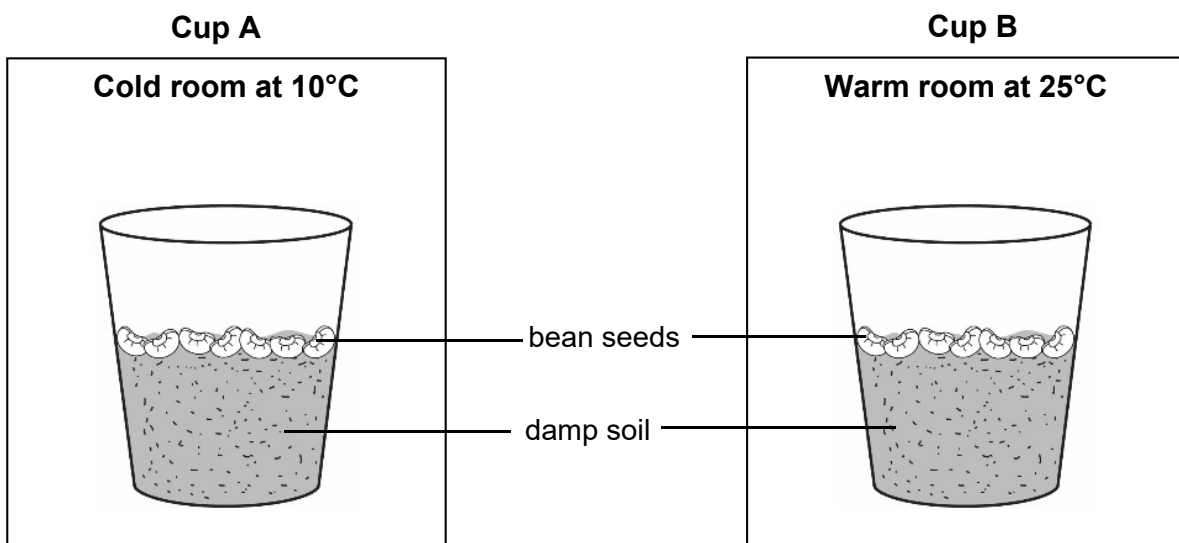
Alia carries out an experiment to investigate how temperature affects seed germination.

On **Day 1**, she places **seven** bean seeds in **each** of two **similar** plastic cups **A** and **B**, as shown in **Diagram 8**.

Both cups contain the **same** amount of **damp** soil.

She then places

- cup **A** in a cold room at 10°C, and
- cup **B** in a warm room at 25°C.



**Diagram 8:** The experiment on seed germination

She records the number of seeds that germinate in each cup from **Day 1** to **Day 5**.

**Table 1** shows her records.

**Table 1:** Number of germinated bean seeds

| Cup      | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|----------|-------|-------|-------|-------|-------|
| <b>A</b> | 0     | 0     | ..... | 2     | 4     |
| <b>B</b> | ..... | 0     | 3     | 5     | ..... |

(a) (i) Complete **Table 1**. [3]

(ii) Which temperature is **more suitable** for bean seed germination? Tick (✓) the correct box.

☐ 10°C

☐ 25°C

[1]

- (iii) Why does Alia keep **all** conditions the **same** in the experiment, **except** temperature?

\_\_\_\_\_

\_\_\_\_\_ [1]

- (b) A suitable temperature is a condition for seed germination. Give **two** other conditions necessary for seed germination.

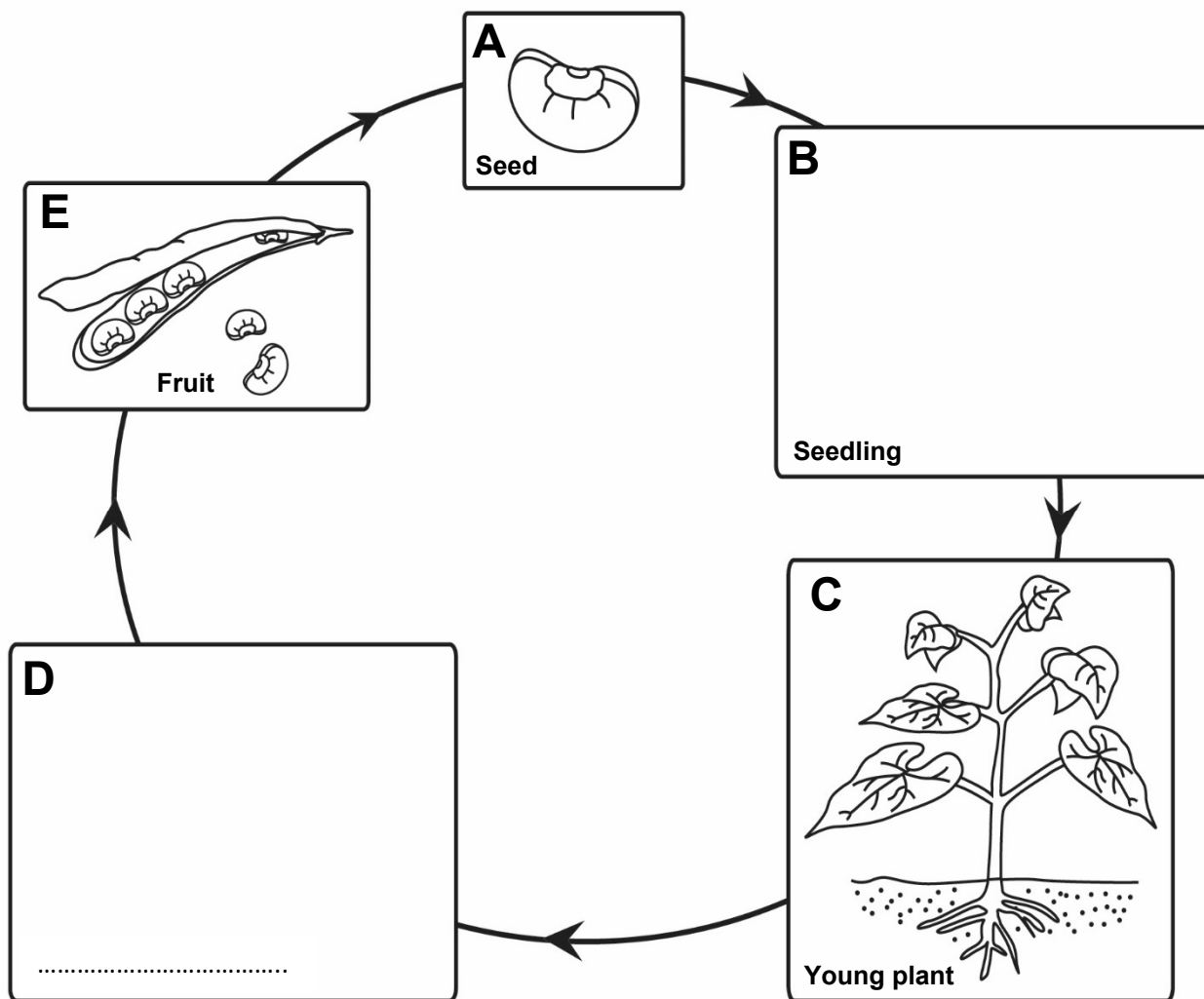
1. \_\_\_\_\_

2. \_\_\_\_\_ [2]

- (c) Parts of the life cycle of a bean plant are shown in **Diagram 9**.

Complete **Diagram 9** by

1. **drawing** the missing parts **B** and **D**;
2. **labelling** part **D**.



**Diagram 9:** Life cycle of a bean plant

[3]

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